

Uzbekistan OS

AI Architecture Specification (AAS)

Version 1.0

1. Purpose

Defines the AI architecture for the Uzbekistan OS MVP, including orchestration, retrieval, prompting, generation, validation, observability, and evaluation.

2. AI Objective

Deliver reliable, personalized, citation-backed answers grounded in verified Uzbekistan knowledge rather than model memory.

3. Design Principles

Retrieval-first generation, evidence-bounded answers, source transparency, structured outputs, model independence, and deterministic business logic.

4. MVP Scope

Support multilingual chat (English, Uzbek, Russian), hybrid retrieval, workflow guidance, citations, personalization, and conversation continuity.

5. High-Level Architecture

User → Chat API → Safety → Query Understanding → Context Builder → Retrieval → Evidence Pack → Answer Generation → Validation → Response.

6. Core Components

Chat API, AI Orchestrator, Retrieval Service, Model Gateway, Prompt Registry, and Evaluation Service.

7. AI Tasks

Language detection, intent classification, entity extraction, query rewriting, answer generation, checklist generation, conversation summarization, and knowledge structuring.

8. Request Lifecycle

Validation, safety checks, context loading, retrieval planning, hybrid retrieval, reranking, evidence construction, answer generation, validation, persistence, and feedback.

9. Query Understanding

Structured schema containing language, intent, entities, retrieval plan, missing information, and risk level.

10. Retrieval Architecture

Hybrid search using metadata filters, PostgreSQL full-text search, pgvector semantic search, applicability filtering, and reranking.

11. Prompt Architecture

Layered prompts composed of platform rules, product rules, domain rules, user context, conversation context, evidence pack, and output schema.

12. Evidence Pack

Compact structured evidence objects with citations, metadata, quality scores, and verified sources.

13. Answer Schema

Structured JSON output including summaries, requirements, steps, warnings, confidence, citations, suggested questions, and workflow recommendations.

14. Conversation Memory

Immediate history, conversation summaries, user profile, and structured context state.

15. Multilingual Support

Native English, Uzbek, and Russian conversations with multilingual retrieval and controlled translation.

16. Model Roles

Classification, embeddings, reranking, answer generation, summarization, and offline knowledge structuring.

17. Safety

Hallucination prevention, prompt-injection defense, structured validation, confidence scoring, and evidence-only generation.

18. Observability

Logging of requests, retrievals, citations, latency, costs, model versions, prompts, and user feedback.

19. Evaluation

Retrieval metrics, answer quality, safety testing, benchmark datasets, human evaluation, and release gates.

20. Final Architecture

Backend-orchestrated retrieval-first AI pipeline designed for reliability, auditability, scalability, and future autonomous capabilities.